

Body Repair Manual

This manual provides clear guidance for inspecting, repairing, and restoring vehicle body components. The goal is to maintain structural integrity, appearance, and safety after collisions or damage.

1. Initial Assessment

- Begin with a visual inspection of all exterior panels, doors, bumpers, and structural areas.
 - Identify dents, scratches, cracks, and deformations.
 - Check for misaligned panels or gaps that could indicate internal damage.
 - Document all findings before starting any repair work.
-

2. Damage Categorization

- **Minor Surface Damage:** Light scratches, paint chips, small dents.
 - **Moderate Damage:** Larger dents, bent trims, minor structural distortions.
 - **Severe Damage:** Frame or subframe misalignment, deep corrosion, or major collision impact.
-

3. Surface Preparation

- Clean the area thoroughly to remove dirt, grease, or debris.
 - Sand or abrade damaged areas as required for paint or filler adhesion.
 - Mask surrounding panels to protect from overspray or accidental scratches.
-

4. Panel Repair Procedures

- **Dent Removal:** Use pulling tools, hammers, or heat-assisted techniques depending on severity.
- **Filler Application:** Apply body filler to smooth irregular surfaces, then sand to match panel contours.
- **Panel Replacement:** For severely damaged panels, remove and replace using approved hardware and alignment tools.

5. Painting and Finishing

- Match paint using factory paint codes to ensure accurate color.
- Apply primer, base coat, and top coat according to manufacturer specifications.
- Allow appropriate drying and curing times between layers.
- Polish and buff the surface to restore factory finish.

6. Structural Checks

- Inspect alignment and mounting points after any repairs.
- Confirm doors, hoods, and tailgates open and close properly without gaps or interference.
- Check suspension and chassis connections if damage was near critical components.

7. Safety Considerations

- Always wear protective gear including gloves, masks, and eye protection.
- Use ventilation when sanding or painting to avoid inhaling fumes.
- Follow manufacturer-approved repair methods to maintain crashworthiness.

8. Documentation and Quality Control

- Record all repairs performed, parts replaced, and materials used.
- Take before-and-after photos to confirm quality and completeness.